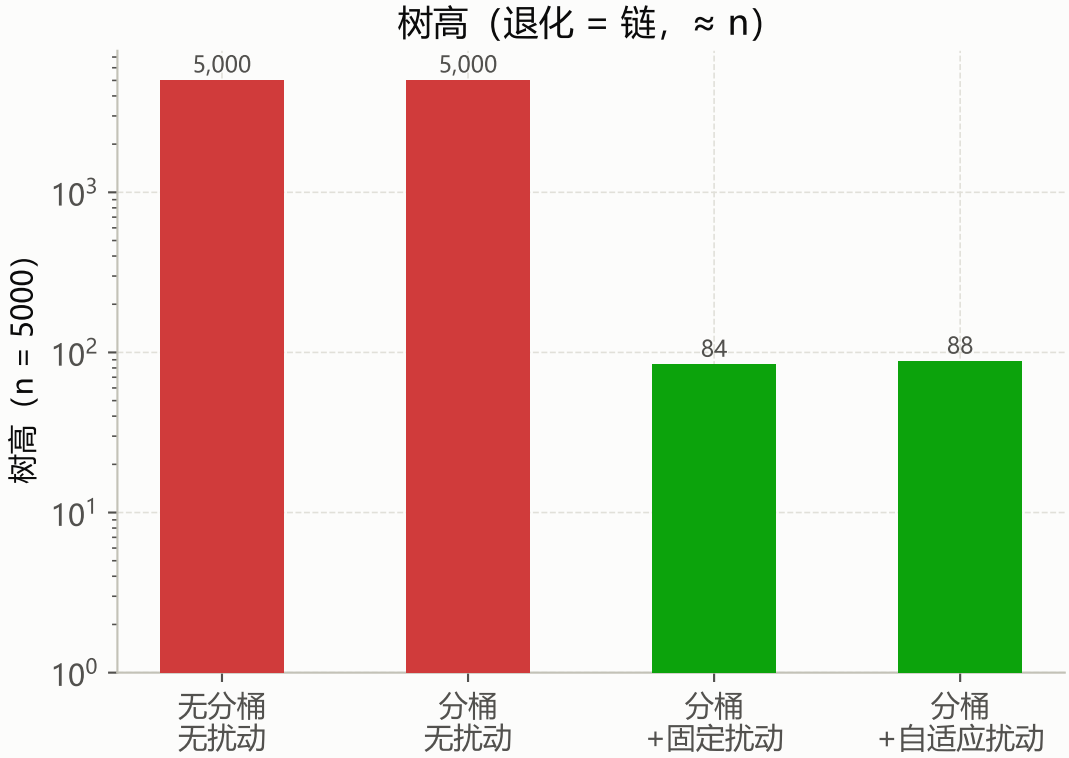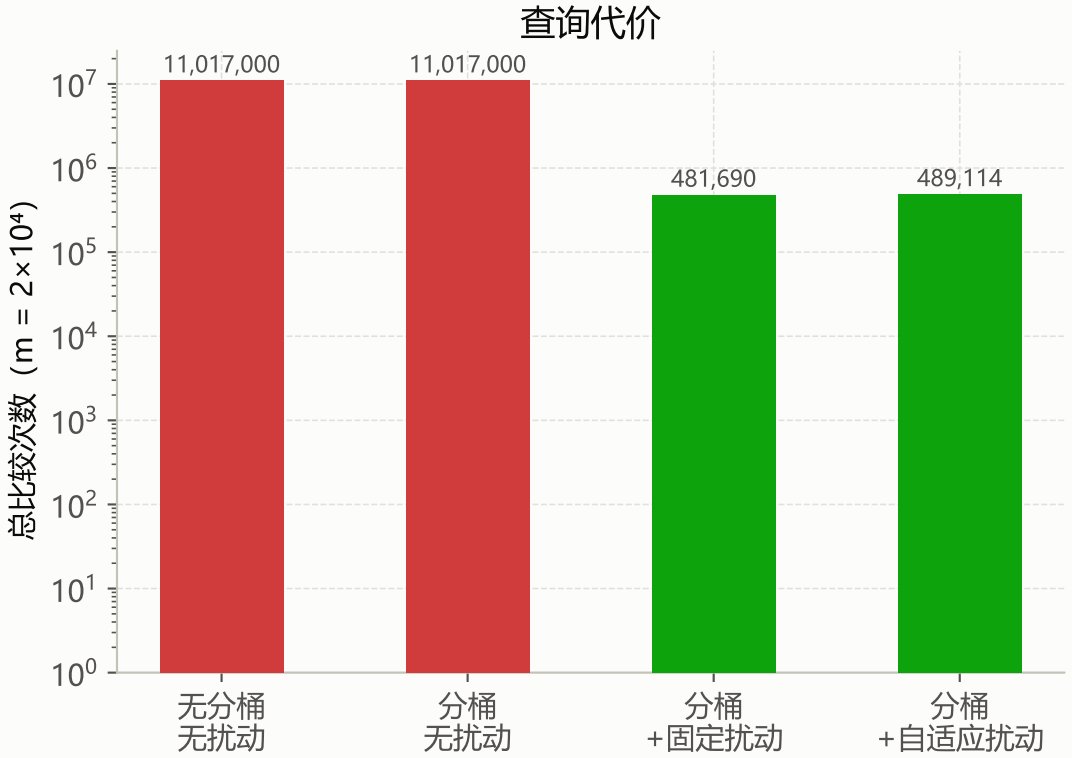图 7　E7　分桶 + 扰动必要性消融（对抗键序，Zipf α=1, n=5000）